

Think — compound figures in the world

Sports pitches, windows, tiles and gardens — real shapes, cut into known pieces.

LESSON 106

1 × 40 MIN

MATEMATIKA 6, O'YLAB KO'R

S7 15 IN CONTEXT

LESSON CLOCK

10'

12'

12'

6'

Modelling a real shape	10 min
Area or length?	12 min
Costing a job	12 min
How accurate?	6 min

LEARNING OBJECTIVES

- Model a real object as a compound of simple shapes.
- Decide whether the question wants an area or a length.
- Estimate a real quantity from a measured plan.
- Judge the accuracy of the answer.

TEXTBOOK

National	pp. 311–313
Cambridge	Stage 7, pp. 148–156
Workbook	Exercise 15.5

01

Explanation

Modelling a real shape

OBJECT	MODELLED AS
a running track	a rectangle with two semicircular ends
an arched window	a rectangle with a semicircle on top
a football pitch	a rectangle, with a circle in the middle
a paved corner	a square less a quadrant
a garden with a round pond	a rectangle less a disc

EVERY MODEL LEAVES SOMETHING OUT

A real track has a kerb, a real window has a frame of some thickness. Saying what has been ignored is part of a good answer, not an apology for it.

Area or length?

WHAT IS WANTED	MEASUREMENT	UNITS
grass seed for a lawn	area	m^2
fencing round it	perimeter	m
glass for a window	area	cm^2
beading round the glass	perimeter	cm
paint for a wall	area	m^2
skirting board	length	m

THE MATERIAL TELLS YOU WHICH

Anything sold by the roll or the metre is a length; anything sold by the tin or the square metre is an area. Read the material, not the shape.

Costing a job

PROBLEM	WORKING	ANSWER
turf at 35 000 sum a m^2 for a 12×8 m lawn	$96 \cdot 35\,000$	3 360 000
fencing at 25 000 a metre round the same lawn	$40 \cdot 25\,000$	1 000 000
the same lawn with a 2 m round pond removed	$96 - 12.57$	83.43 m^2
tiles 20 cm square for a 3×4 m floor	$12 \div 0.04$	300 tiles

CONVERT UNITS BEFORE DIVIDING

A 20 cm tile is 0.04 m^2 , not 20 m^2 . Mixing centimetres and metres in one division is where a factor of ten thousand goes missing.

How accurate?

ASSUMPTION	EFFECT
the corners are exactly square	small
$\pi = 3.14$ rather than more places	tiny
no waste when cutting tiles	can be 10%
the ground is flat	can be large on a slope

ADD FOR WASTE, THEN ROUND UP

Nobody orders 83.43 m² of turf. The professional answer is “about 90 m², allowing for waste” — and saying so is a mathematical judgement, not a vague one.

02 Worked examples

EXAMPLE 1 A lawn is 12 m by 8 m with a circular pond of radius 2 m in it. Find the grassed area.

1

Rectangle: 96 m².

2

Pond: $3.14 \cdot 4 = 12.57$ m².

3

$96 - 12.57 = 83.43$ m².

4

Order about 90 m² of turf, allowing for waste.

ANSWER

83.43 m²

EXAMPLE 2 How many 20 cm square tiles cover a floor 3 m by 4 m?

- ① Floor: 12 m^2 .
- ② One tile: $0.2 \cdot 0.2 = 0.04 \text{ m}^2$.
Convert first.
- ③ $12 \div 0.04 = 300$ tiles.

ANSWER

300 tiles

EXAMPLE 3 An arched window is a 80 cm by 120 cm rectangle with a semicircle of radius 40 cm on top. Find the glass area.

- ① Rectangle: $9\,600 \text{ cm}^2$.
- ② Semicircle: $\frac{1}{2} \cdot 3.14 \cdot 1\,600 = 2\,512 \text{ cm}^2$.
- ③ $9\,600 + 2\,512 = 12\,112 \text{ cm}^2$.
About 1.2 m^2 .

ANSWER $12\,112 \text{ cm}^2$ **03 Interactive model**

Measure a real room or courtyard and cost a job in it; the arithmetic is the same as the textbook and the answer can be checked against a real price.

Area, length, or number of pieces?

Grass seed for a lawn is bought by:

the metre

the square metre

the piece

the litre

Fencing is bought by:

the metre

the square metre

the tin

the tile

A 12×8 m lawn has area:

40

96

20

192

Its perimeter is:

20

40

96

48

A pond of radius 2 m has area about:

6.28

12.57

4

25.1

A 20 cm square tile is:

20 m²

0.4 m²

0.04 m²

400 m²

Tiles for a 12 m² floor:

60

300

600

30

Ordering exactly the calculated area is:

sensible

risky — allow for waste

required

impossible

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Model	Model	Модель
Plan	Reja	План
Estimate	Baholash	Оценка
Material	Material	Материал
Cost	Narx	Стоимость
Per square metre	Bir kvadrat metrga	За квадратный метр
Accuracy	Aniqlik	Точность
Assumption	Faraz	Допущение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A running track is modelled as:

a circle

a rectangle with semicircular ends

a square

an oval with no formula

Paint is bought by:

the metre

the area to cover

the perimeter

the tile

Skirting board is bought by:

the area

the length

the piece

the tin

A 20 cm tile covers:

0.04 m²

0.2 m²

4 m²

20 m²

A model of a real shape:

is exact

ignores some details

is useless

needs no assumptions

A sensible order for 83.43 m² of turf is:

83.43

about 90

80

83

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. A 12×8 m lawn: the area

ANSWER 96 m^2

2. Its perimeter

ANSWER 40 m

3. Turf at $35\,000$ a m^2

ANSWER $3\,360\,000$ sum

4. Fencing at $25\,000$ a metre

ANSWER $1\,000\,000$ sum

5. A 20 cm tile in m^2

ANSWER 0.04

6. Tiles for 12 m^2

ANSWER 300

7. Grass seed is measured by

ANSWER area

Medium · the standard question

7 PROBLEMS

1. A 12×8 lawn with a pond of radius 2 m

ANSWER 83.43 m^2

2. An arched window 80×120 with a semicircle of radius 40

ANSWER $12\,112 \text{ cm}^2$

3. A 3×4 m floor in 25 cm tiles

ANSWER 192 tiles

4. Paint covering 10 m^2 a litre for 45 m^2

ANSWER 4.5 litres

5. A 5×4 m room: skirting board

ANSWER 18 m

6. A track 80×70 with ends of radius 35: the area

ANSWER 9450 m^2

7. Which needs the perimeter: turf or fencing?

ANSWER Fencing

Hard · stretch and reason

7 PROBLEMS

1. A lawn 20×15 m with two round beds of radius 3 m

ANSWER 243.5 m^2

2. Turf for that lawn at 40 000 a m^2 , allowing 10% waste

ANSWER About 10 700 000 sum

3. A room 6×4 m with a door 0.9 m wide: the skirting

ANSWER 19.1 m

4. A wall 8 m by 2.5 m with a 2×1.5 m window: the paint area

ANSWER 17 m^2

5. Why allow for waste?

ANSWER Cut pieces at the edges cannot always be reused

6. A 30 cm tile for the same 12 m^2 floor

ANSWER 134 tiles, rounded up

7. State two assumptions in the lawn problem

ANSWER The ground is flat and the corners are square

07 Homework

Homework — 5 tasks

Say what you are assuming, and round the final order sensibly.

1. Measure a room at home and find its floor area and perimeter.
2. Work out how many 30 cm square tiles would cover that floor.
3. A lawn 15×10 m has a circular pond of radius 2.5 m. Find the grassed area.
4. Cost turf for that lawn at 38 000 sum a square metre, allowing 10% for waste.
5. Write two sentences saying what your model of the lawn has ignored.